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2D Virtual Try-On Accuracy Improvement Documentation

1. Introduction

The 2D Virtual Try-On system aims to digitally fit garments onto users while preserving body structure, clothing appearance, and realism. The main objective of this optimization phase is to improve try-on quality by enhancing clothing alignment, segmentation accuracy, pose estimation, garment overlay quality, and edge smoothness.

This work focuses on reducing visual artifacts, improving fitting accuracy, and increasing realism for various user poses and body structures. The optimization process includes testing multiple poses, improving human parsing modules, refining clothing masks, and handling challenging backgrounds.

2. Objectives

The primary objectives of this optimization work are:

- Improve clothing alignment accuracy
- Enhance body segmentation quality
- Improve pose detection precision
- Achieve realistic garment overlay
- Reduce boundary artifacts using smoothing techniques
- Handle different body poses effectively
- Improve background separation quality

3. Current 2D Try-On Workflow

Current workflow followed in the virtual try-on system:

User Image + Garment Image
↓
Image Preprocessing
↓
Pose Detection
↓
Human Parsing / Segmentation
↓
Garment Warping
↓

Garment Overlay
↓
Rendering & Smoothing
↓
Final Try-On Output

4. Clothing Alignment Optimization

Objective

Improve garment fitting and positioning accuracy over the human body.

Existing Issue

Observed problems:

• Sleeve misalignment

• Shoulder mismatch

• Incorrect torso fitting

• Neck overlap

• Garment stretching artifacts

Optimization Methods

The following methods were explored:

TPS (Thin Plate Spline) Warping

TPS warping deforms garment images according to body posture and body landmarks.

Advantages:

• Better sleeve fitting

• Flexible garment transformation

• Reduced deformation artifacts

DensePose Integration

DensePose maps human body surfaces and improves garment placement.

Benefits:

• Better body surface understanding

• Improved alignment accuracy

• Better body-to-cloth correspondence

OpenPose Landmark Guidance

OpenPose extracts body landmarks:

- Shoulder points

- Elbows
- Wrists
- Torso landmarks
- Hip locations

These landmarks guide garment positioning.

Testing Scenarios

Garment fitting was tested using various poses:

Test Case	Description
Pose 1	Front standing
Pose 2	Slight side pose
Pose 3	Arms raised
Pose 4	Bent arm pose
Pose 5	Loose clothing pose

Evaluation metrics:

• Sleeve fitting quality

• Shoulder alignment

• Neck positioning

• Torso coverage

5. Human Parsing Optimization

Objective

Improve body segmentation quality and garment layering.

Existing Challenges

Problems identified:

• Arm segmentation errors

• Hair overlap

• Background leakage

• Torso masking issues

Detectron2 Integration

Detectron2 improves instance segmentation quality.

Capabilities:

• Body separation

• Background removal

• Better clothing masks

• Improved region extraction

SCHP Parsing

Self-Correction Human Parsing (SCHP) improves human body segmentation.

Segmented regions:

• Face

• Hair

• Arms

• Torso

• Legs

• Background

Benefits:

• Better garment layering

• Reduced occlusion errors

• Improved realism

Evaluation Table

Component	Status
Arm Detection	Tested
Hair Separation	Tested
Torso Masking	Tested
Background Isolation	Tested

6. Pose Detection Optimization

Objective

Improve body landmark accuracy for realistic fitting.

MediaPipe Pose Detection

MediaPipe extracts lightweight body landmarks.

Advantages:

• Fast inference

• Real-time processing

• Low latency

Detected points:

- Shoulder
- Elbow
- Wrist
- Hip
- Knee

OpenPose Integration

OpenPose improves fitting precision.

Benefits:

• Multi-point detection

• Better sleeve alignment

• Accurate torso fitting

• Better pose robustness

Pose Testing

Different poses evaluated:

1. Normal standing
2. Side orientation
3. Raised arms
4. Walking posture
5. Hand movement variations

7. Garment Overlay Optimization

Objective

Increase realism and reduce overlay artifacts.

Existing Problems

Observed artifacts:

• Floating garments

• Color mismatch

• Layer inconsistency

• Unrealistic fitting

GAN-Based Rendering

GAN rendering improves:

• Visual realism

• Texture consistency

• Cloth appearance

• Surface blending

Benefits:

• Natural appearance

• Reduced artifacts

• Better texture retention

8. Edge Smoothing Optimization

Objective

Reduce visible boundaries around garments.

Existing Problems

Issues observed:

• Jagged borders

• Hard edges

• Sharp transitions

Optimization Techniques

Alpha Blending

Improves transition between garment and body.

Feather Masking

Softens garment boundaries.

Gaussian Edge Smoothing

Reduces visible artifacts.

Result:

Improved garment continuity and realism.

9. Background Handling Optimization

Objective

Reduce background noise affecting segmentation.

Problems

Detected issues:

• Background leakage

• Object interference

• Segmentation noise

Portrait Matting

Portrait matting isolates users from backgrounds.

Benefits:

• Cleaner body extraction

• Improved overlay quality

• Better segmentation

Background Testing Cases

Scenario	Evaluation
Plain background	Tested
Indoor scene	Tested
Outdoor scene	Tested
Complex background	Tested

10. Challenges Encountered

Major challenges observed:

Clothing Related

• Loose garments

• Cloth deformation errors

• Sleeve mismatch

Human Parsing Related

• Hair occlusion

• Arm overlap

• Incorrect masks

Pose Related

• Side poses

• Dynamic movement

• Landmark shifts

Environment Relate

• Lighting variation

• Complex backgrounds

• Noise

11. Future Improvements

Future optimization plans:

• HR-VITON integration

• Better DensePose fitting

• Improved cloth simulation

• Real-time optimization

• Better segmentation models

• Multi-angle fitting support

12. Conclusion

The optimization process significantly improved the 2D virtual try-on system by enhancing garment alignment, human parsing, pose estimation, garment overlay quality, and edge smoothing.

The improvements increased fitting realism, reduced visual artifacts, and improved robustness across different body poses and environments.

Future work will focus on advanced garment deformation methods and real-time deployment optimization.